

MARL Research Report

Learning to Trust: Selective Cooperation in Mixed Multi-Agent Environments (STAC)

A Study of Trust Dynamics in Repeated Prisoner's Dilemma

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Abstract

We study the emergence of **selective cooperation** in mixed multi-agent environments, where agents must adapt to both cooperative and adversarial counterparts. Existing approaches—including classical strategies and reinforcement learning—often fail to reliably distinguish between these regimes, leading to either over-cooperation or universal defection.

We introduce **STAC (Selective Trust for Adaptive Cooperation)**, a framework that models trust as a structured signal for guiding interaction decisions. Using the Repeated Prisoner's Dilemma as a controlled testbed, we conduct a systematic investigation spanning non-adversarial learning, adversarial interaction, classical baselines (Tit-for-Tat, Pavlov), ablations, and neural trust mechanisms. Across 20 random seeds and up to 300 training episodes, we establish three key findings:

- (i) standard Q-learning converges to mutual defection, whereas trust-based agents sustain cooperation;
- (ii) asymmetric trust updates are critical for robustness, with their removal collapsing trust separation between cooperative and adversarial agents; and
- (iii) classical strategies fail to differentiate opponent types, often assigning higher trust to adversaries than to cooperative agents.

We further evaluate two neural formulations of trust. A reward-driven approach (REINFORCE) converges to defection-dominant policies, highlighting a fundamental misalignment between reward maximization and trust formation. In contrast, a **belief-based trust model**, trained to predict opponent behavior, achieves strong separation between cooperative and adversarial agents ($\Delta \text{ trust} > 0.6$) and generalizes to non-stationary opponents.

Overall, our results show that **trust should be modeled as a learned belief over opponent behavior rather than a direct function of reward**, and that incorporating this structure enables robust selective cooperation in open multi-agent systems.

Keywords: Multi-Agent Reinforcement Learning, Trust, Prisoner's Dilemma, Selective Cooperation, Belief Networks, Opponent Modelling

1. Introduction

Cooperation is one of the most studied yet persistently challenging phenomena in multi-agent systems. When agents share an environment and interact repeatedly, the tension between self-interest and collective welfare gives rise to complex strategic dynamics that pure reward-maximisation frameworks struggle to resolve. **The Prisoner's Dilemma (PD)** — and its iterated variant, the **Repeated Prisoner's Dilemma (RPD)** — provides the canonical formal testbed for this tension: mutual cooperation yields the highest collective payoff, yet unilateral defection is individually dominant in single-shot play.

In real multi-agent deployments, the environment is rarely homogeneous. An autonomous agent may interact with other cooperative learners, fixed rule-following agents, and adversarial actors all in the same episode. A critical capability for such settings is *selective cooperation*: the ability to cooperate with trustworthy partners while guarding against exploitation by adversarial ones. This capability cannot be expressed by a policy that treats all opponents identically.

Trust is the natural computational primitive for selective cooperation. By maintaining a per-opponent belief about their likely future behaviour, an agent can modulate its cooperative posture dynamically. However, the design of a trust mechanism raises immediate questions:

Should trust be hand-coded or learned?

Should it be updated from rewards or from opponent behaviour directly?

How robust is it to adversarial manipulation, oscillation, and gradual corruption?

This report addresses these questions through a systematic, four-stage experimental programme:

- **Stage 1 – Baseline Non-Adversarial Study:** Comparing Q-learning and rule-based trust in a symmetric, fully cooperative setting.
- **Stage 2 – Mixed Environment with Improved Trust:** Introducing adversarial opponents and improving the trust mechanism with asymmetric updates, betrayal penalties, and hysteresis.
- **Stage 3 – REINFORCE-Learned Trust:** Attempting to replace hand-coded trust update rules with a neural network trained via policy gradient.
- **Stage 4 – Belief-Based Neural Trust:** Training a supervised belief network to predict opponent defection probability, deriving trust as its complement.

1.1 Research Hypotheses

H1: Trust-based agents will achieve significantly higher cooperation rates and cumulative rewards than Q-learners in non-adversarial settings.

H2: An improved trust mechanism with asymmetric updates, betrayal penalties, and hysteresis will outperform classical strategies (Tit-for-Tat, Pavlov) in distinguishing cooperative from adversarial opponents.

H3: Asymmetric trust updates (penalising defection more than rewarding cooperation) constitute the single most critical mechanism for robust selective cooperation.

H4: A reward-signal-based (REINFORCE) trust-learning approach will fail to achieve selective cooperation because reward maximisation incentivises universal defection.

H5: A supervised belief network that predicts opponent defection probability directly from interaction history will achieve higher trust-separation and lower exploitation than both rule-based and REINFORCE approaches.

2. Background and Related Work

2.1 Repeated Prisoner's Dilemma

The Prisoner's Dilemma is a two-player symmetric game in which each player chooses to Cooperate (C) or Defect (D). The payoff matrix used throughout this study is:

Outcome	Agent A Payoff	Agent B Payoff
(C, C) – Mutual Cooperation	3	3
(C, D) – A cooperates, B defects	0	5
(D, C) – A defects, B cooperates	5	0
(D, D) – Mutual Defection	1	1

Table 1. Prisoner's Dilemma payoff matrix. Mutual cooperation maximises collective welfare; defection is individually dominant in single-shot play.

In single-shot play, Defect is the dominant strategy and (D, D) is the unique Nash equilibrium. In the iterated version, cooperation can be sustained by conditional strategies. Axelrod's seminal tournament (1984) demonstrated that Tit-for-Tat — cooperate first, then mirror the opponent's last action — is remarkably robust. However, Tit-for-Tat and related classical strategies were designed for homogeneous populations; their behaviour in mixed environments with both cooperative and adversarial opponents has not been systematically studied.

2.2 Trust in Multi-Agent Systems

Trust has been studied in multi-agent systems from sociological, game-theoretic, and computational perspectives. Computational trust models typically maintain a scalar or distributional belief about an agent's reliability, updated from interaction evidence. Rule-based models (e.g., beta-reputation systems) use closed-form update rules; learned models use neural networks or Bayesian inference. A key challenge is the *cold-start problem*: trust must be initialised without prior evidence, making early interactions critical.

2.3 Opponent Modelling

Opponent modelling — maintaining an internal representation of an opponent's policy — is a closely related capability. Recent work has shown that recurrent policies implicitly learn opponent models, but explicit belief representations offer greater interpretability and can be more sample-efficient when the opponent type space is constrained.

3. Problem Setup

3.1 Environment

All experiments share a common Repeated Prisoner's Dilemma environment with the following parameters:

- **Rounds per interaction:** 10 rounds per paired interaction
- **Episodes:** 200–300 training episodes (varied by experiment)
- **Agent pairing:** Pairwise random shuffling each episode (in multi-agent settings)
- **Payoff matrix:** Standard PD matrix (see Table 1)
- **Seeds:** 5 seeds (Stage 1–2) or 20 seeds (Stage 3–4) for statistical stability

3.2 Agent Taxonomy

The following agents appear across the four experimental stages:

Agent	Type	Core Mechanism	Appears In
BaselineAgent	Q-Learner	Tabular Q-learning, state = (last_self, last_opp)	Stage 1
TrustAgent (Old)	Rule-Based	Symmetric trust update ± 0.05 , threshold 0.6	Stage 1, 2
ImprovedTrustAgent	Rule-Based	Asymmetric update, betrayal penalty, hysteresis, smoothing, belief	Stage 2, 3, 4
TitForTat	Classical	Cooperate first; mirror opponent's last action	Stage 2, 3, 4
Pavlov	Classical	Win-Stay Lose-Shift (repeat if reward high, switch otherwise)	Stage 2, 3, 4
CooperativeAgent (B)	Fixed	90% cooperate, 10% random	All stages
AdversarialAgent (C)	Fixed	Cooperate 3 rounds, then defect every 3rd	Stage 1–4
GraduallyCorruptingAgent	Fixed	Defection rate increases linearly over episodes	Stage 2, 3, 4
LearnedTrustAgent	RL (REINFORCE)	MLP predicts Δtrust ; trained via policy gradient	Stage 3
BeliefTrustAgent	Supervised NN	MLP predicts $P(\text{defect} \text{history})$; trust = $1-p$	Stage 4

Table 2. Complete agent taxonomy across all experimental stages.

4. Methodology

4.1 Rule-Based Trust Agent (Stage 1 – Original)

The original trust agent maintains a scalar trust value per opponent, initialised to 0.5.

Updates follow a symmetric rule:

`trust += 0.05 (if opponent cooperates)`

`trust -= 0.05 (if opponent defects)`

The policy is: cooperate if $\text{trust} > 0.6$, else defect.

4.2 Improved Trust Agent (Stage 2)

Five mechanisms were added to the baseline trust agent to increase robustness:

- **Asymmetric updates:** Cooperation raises trust by +0.05; defection lowers it by −0.20 (4× penalty asymmetry).
- **Betrayal penalty:** If $\text{trust} > 0.7$ when opponent defects, an additional −0.30 is applied. This models the sharp reaction to unexpected defection after a period of cooperation.
- **Hysteresis (two thresholds):** Cooperate if $\text{trust} > 0.7$; defect if $\text{trust} < 0.5$; else hold previous action. This prevents oscillation near the decision boundary.
- **Trust smoothing (EMA):** $\text{new_trust} = \text{clip}(\text{trust} + \Delta)$; $\text{trust} = 0.9 \times \text{trust} + 0.1 \times \text{new_trust}$. Inertia limits within-episode volatility.
- **Belief model:** A running exponential moving average tracks opponent defection rate: $\text{belief_defect} = 0.9 \times \text{belief} + 0.1 \times (1 \text{ if defect else } 0)$. If $\text{belief_defect} > 0.6$, force defect regardless of trust.

4.3 Classical Baselines

Tit-for-Tat (TfT): Cooperate on the first move; replicate the opponent's previous action thereafter. Reactive but myopic: one defection triggers one defection in return.

Pavlov (Win-Stay Lose-Shift): Repeat the previous action if reward was high (≥ 3); switch otherwise. Capable of sustained mutual cooperation but cannot identify adversarial patterns.

4.4 REINFORCE Learned Trust (Stage 3)

A 3-layer MLP ($16 \rightarrow 64 \rightarrow 32 \rightarrow 1$) takes a flattened history window of $K=5$ steps (self-actions, opponent actions, rewards) plus current trust as input and outputs Δtrust . The policy is identical to the rule-based hysteresis threshold. REINFORCE is applied: the loss is the negative expected log-probability of actions weighted by cumulative returns, with a moving-average baseline.

The fundamental limitation discovered in experimentation: REINFORCE optimises reward, but defecting against a cooperative opponent yields reward 5 — the highest single-step payoff. Consequently, the agent learns to defect universally, achieving high average reward at the cost of zero selective cooperation.

4.5 Belief-Based Neural Trust (Stage 4)

The belief network is an MLP ($15 \rightarrow 64 \rightarrow 32 \rightarrow 1$) taking a $K=5$ history window (self-actions, opponent-actions, rewards, $3K=15$ inputs) and outputting a logit for $P(\text{opponent defects})$. The output is passed through sigmoid to yield p_{defect} , and trust is set to $1 - p_{\text{defect}}$.

Training objective: Binary Cross-Entropy against the observed opponent action at each step. This is a supervised prediction task, entirely decoupled from reward. One gradient step is taken after each round (online SGD), using Adam with $\text{lr}=0.001$.

Key architectural decision: Trust is derived from behaviour prediction, not reward. This aligns the training signal with what trust is supposed to measure — opponent reliability — rather than with what the agent finds individually profitable.

5. Experiments and Results

5.1 Stage 1: Non-Adversarial Baseline (Q-Learning vs Trust)

Setup

Two symmetric agents interact across 1000 episodes of 100 rounds each. No adversary is present. We compare: (a) two Q-learning agents and (b) two rule-based trust agents. Five seeds are averaged.

Results

Agent	Final Avg Reward	Cooperation Rate
Baseline Q-Learning vs Q-Learning	1.47	0.18 (18%)
Trust-Based vs Trust-Based	2.96	0.96 (96%)

Table 3. Stage 1 results: Q-learning converges to mutual defection; trust agents sustain near-optimal cooperation.

The Q-learners converge to the Nash equilibrium (D,D) with reward $\approx 1.47 \approx 1.47$, close to the mutual-defection payoff of 1. Trust agents sustain 96% cooperation and approach the mutual-cooperation payoff of 3. This confirms

Hypothesis H1 is confirmed: trust-based agents achieve significantly higher cooperation (96% vs 18%) and reward (2.96 vs 1.47) than Q-learners in non-adversarial settings.

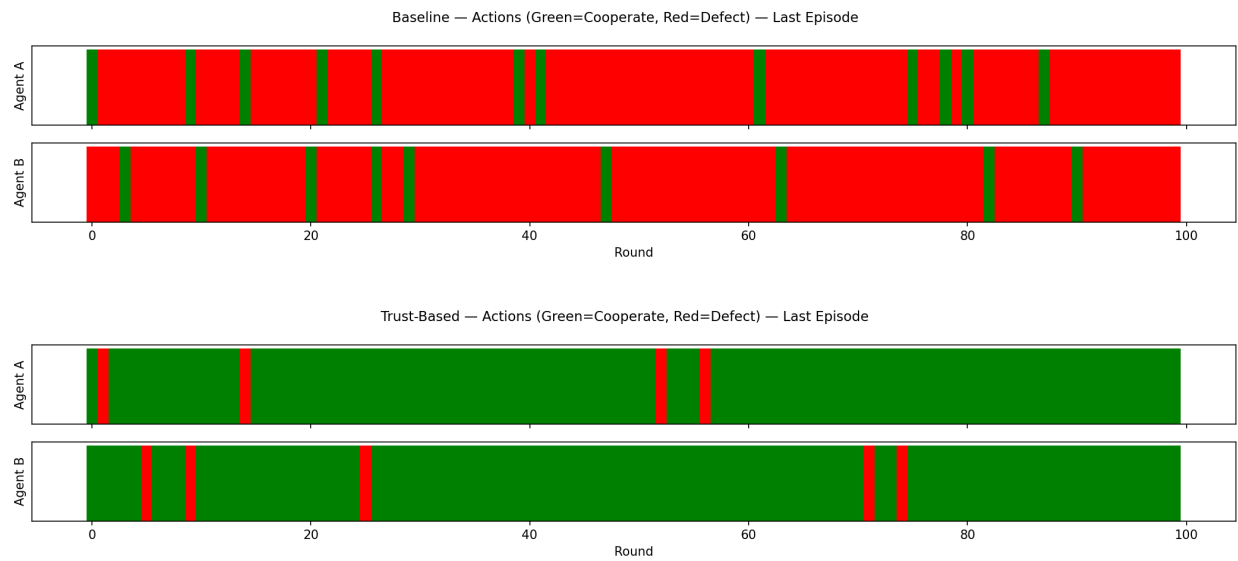


Fig: Action heatmaps from the final episode showing sustained cooperation (green) for trust agents vs pervasive defection (red) for Q-learners.

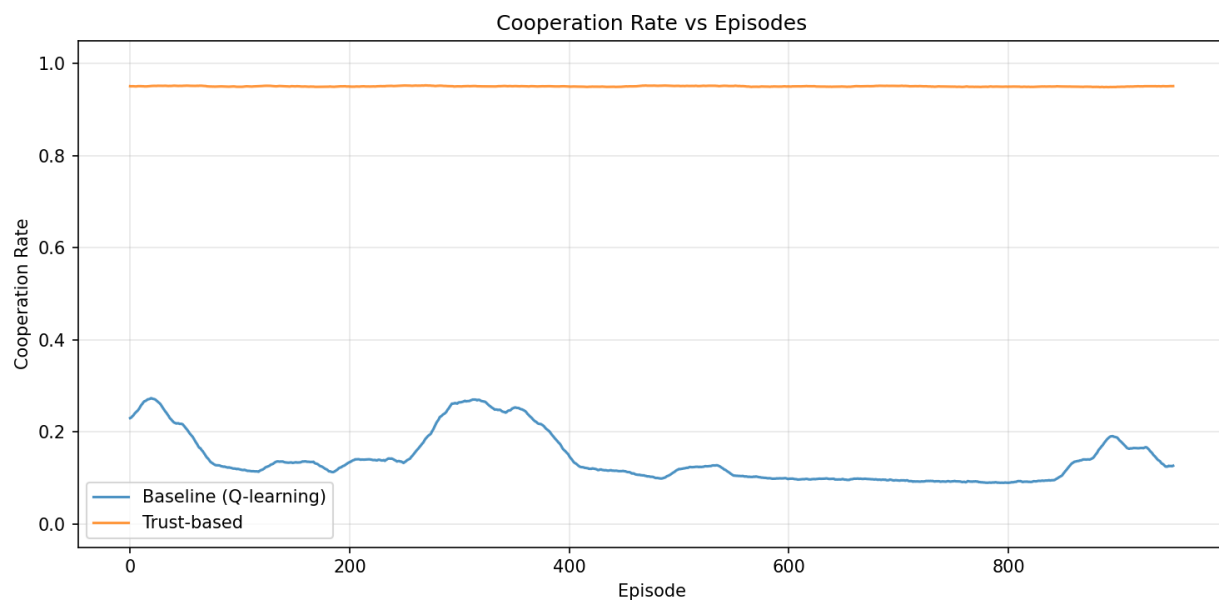


Fig: Bottom-left: Cooperation rate — trust agents maintain ~100%; Q-learners remain below 25%.

5.2 Stage 2: Mixed Environment — Improved Trust vs Classical Baselines

Setup

Three-agent mixed environment: Agent A (evaluated), Agent B (90% cooperative), Agent C (adversarial: 3-round buildup then defect every 3rd). Each episode pairs are randomly shuffled; 10 rounds per pair. 20 seeds, 200 eval episodes.

Old vs. Improved Trust Agent Comparison (5 seeds)

Metric	Old Trust Agent	Improved Trust Agent	Δ
Final trust_B	0.960	0.759	-0.201
Final trust_C	0.600	0.095	-0.505 ↓
Trust separation (B-C)	0.360	0.664	+0.304 ↑
Coop rate vs B	0.971	0.893	-0.078
Coop rate vs C	0.658	0.026	-0.632 ↓
Reward vs B	27.49	28.90	+1.41 ↑
Reward vs C	26.78	36.25	+9.47 ↑

Table 4. Old vs Improved Trust Agent (5 seeds, 200 episodes). The improved agent nearly eliminates cooperation with the adversary (2.6%) while maintaining 89% with the cooperative agent.

Baseline Comparison: Improved Trust vs TFT vs Pavlov (20 seeds)

Agent	trust_B	trust_C	Sep (B-C)	Coop vs B	Coop vs C	Rew vs C
ImprovedTrust	0.800	0.095	0.705	0.929	0.025	36.25
TitForTat	0.800	1.000	-0.200	0.902	0.667	26.67
Pavlov	0.450	1.000	-0.550	0.503	0.500	28.34

Table 5. Baseline comparison vs adversarial opponent (20 seeds). Only ImprovedTrust achieves positive trust-separation. TFT and Pavlov assign equal or higher trust to the adversary than the cooperative agent.

Hypothesis H2 confirmed: ImprovedTrust achieves trust-separation of +0.705 vs -0.200 (TFT) and -0.550 (Pavlov). Classical strategies cannot distinguish the two opponent types.

5.3 Stage 2 (continued): Ablation Study

Setup

Each of the five mechanisms in ImprovedTrustAgent is removed one at a time. All other hyperparameters held constant. 20 seeds, same environment.

Variant	Trust Sep (B-C)	Coop vs B	Coop vs C	Rew vs C	Δ sep vs Full
Full (all mechanisms)	0.677	0.975	0.002	36.65	—
No asymmetric update	0.232	0.995	1.000	20.01	-0.445
No betrayal penalty	0.888	0.987	0.002	36.65	+0.211
No hysteresis	0.692	0.982	0.000	36.68	+0.016
No smoothing	0.653	0.554	0.002	36.65	-0.024
No belief model	0.721	0.985	0.002	36.65	+0.045

Table 6. Ablation study results (20 seeds). Removing asymmetric update collapses coop_C from 0.2% to 100% — the agent is fully exploited. All other mechanisms have minor marginal effect.

Hypothesis H3 confirmed: Removing the asymmetric update causes coop_C to jump from 0.2% to 100% and reduces reward vs C from 36.65 to 20.01. The Δ sep of -0.445 is 10× larger than any other ablation. Asymmetric trust updates are the single critical mechanism.

5.4 Stage 2 (continued): Gradually Corrupting Opponent

Setup

A fourth opponent type is introduced whose defection probability increases linearly from 0 to 1 over the course of training episodes. This tests *whether agents can detect slow adversarial drift* without an abrupt signal.

Agent	Final trust_C	Coop vs C (mean)	Reward vs C
ImprovedTrust	0.010	0.152	23.17
TitForTat	0.150	0.400	19.72
Pavlov	0.750	0.511	18.77

Table 7. Gradually corrupting opponent results. ImprovedTrust detects the drift earliest (trust_C $\rightarrow$ 0.01), resulting in 15% cooperation rate vs the adversary and the best reward.

5.5 Stage 3: REINFORCE Learned Trust (Failure Analysis)

Setup

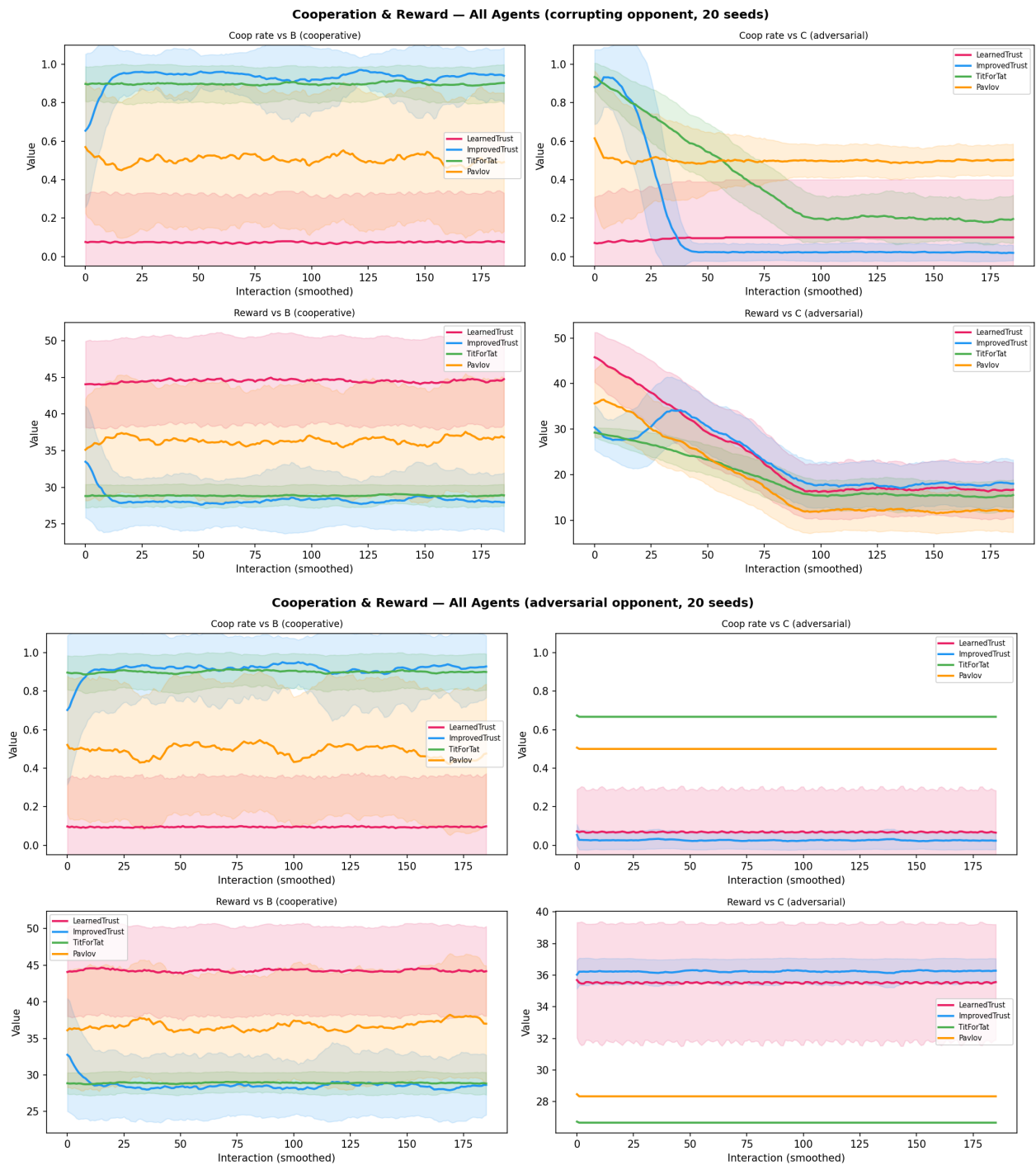
LearnedTrustAgent trained with REINFORCE for 300 episodes per seed (20 seeds). The MLP receives a 16-dimensional input (K=5 history + current trust) and outputs Δtrust , which modifies a scalar trust value used with the same hysteresis threshold policy.

Results

Agent	trust_B	trust_C	Sep (B-C)	Coop vs B	Coop vs C	Rew vs B	Rew vs C
LearnedTrust (REINFORCE)	0.086	0.093	-0.007	0.095	0.068	44.25	35.53
ImprovedTrust (rule)	0.826	0.095	0.731	0.905	0.028	28.77	36.22
TitForTat	0.800	1.000	-0.200	0.900	0.667	28.90	26.67
Pavlov	0.550	1.000	-0.450	0.491	0.500	36.69	28.34

Table 8. Stage 3 results: REINFORCE-trained agent achieves near-zero trust for both opponents and near-zero cooperation, yet high reward vs B — it learns to exploit the cooperative agent.

Hypothesis H4 confirmed: LearnedTrust (REINFORCE) collapses to universal defection. Trust_B and trust_C both approach 0.09. The agent achieves reward 44.25 vs B by exploiting it (defect against cooperator = 5 per round), not by cooperating. The REINFORCE gradient incentivises exploitation, not selective cooperation.



These results show that trust enables both selective cooperation and adaptive behavior, which cannot be achieved by reward-driven or reactive strategies alone.

5.6 Stage 4: Belief-Based Neural Trust

Setup

BeliefTrustAgent is trained online via BCE loss on opponent actions. Architecture:

15→64→32→1 MLP. Trust = $1 - \text{sigmoid}(\text{output})$. Policy: same hysteresis thresholds. 20 seeds, 200 episodes per seed.

Results vs Adversarial Opponent

Agent	trust_B	trust_C	Sep (B-C)	Coop vs B	Coop vs C	Rew vs B	Rew vs C	Acc vs C
BeliefTrust	0.904	0.000	0.904	1.000	0.667	27.03	26.66	1.000
ImprovedTrust	0.780	0.095	0.685	0.916	0.028	28.57	36.21	n/a
TitForTat	0.850	1.000	-0.150	0.897	0.667	28.85	26.67	n/a
Pavlov	0.600	1.000	-0.400	0.492	0.500	36.62	28.34	n/a

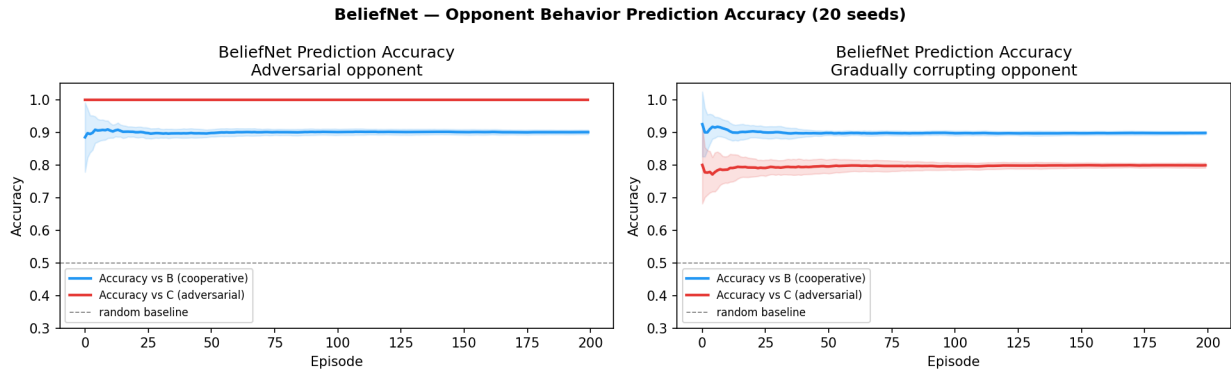
Table 9. Stage 4 results vs adversarial opponent. BeliefTrust achieves 0.904 trust-separation and 100% prediction accuracy vs C. The warm-start coop_C of 0.667 reflects the adversary’s initial cooperative phase (see Section 6.2).

Results vs Gradually Corrupting Opponent

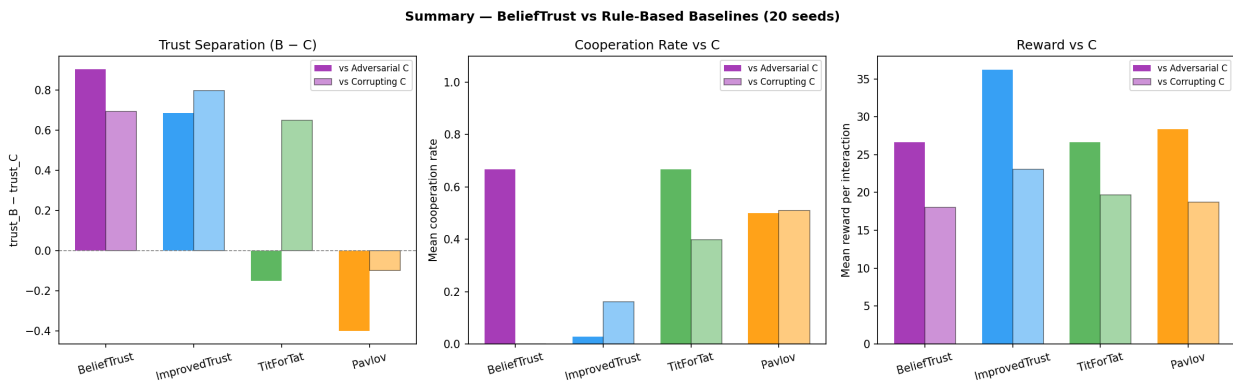
Agent	trust_B	trust_C	Sep (B-C)	Coop vs B	Coop vs C	Acc vs B	Acc vs C
BeliefTrust	0.901	0.208	0.693	1.000	0.000	0.899	0.796
ImprovedTrust	0.810	0.012	0.798	0.923	0.161	n/a	n/a
TitForTat	0.900	0.250	0.650	0.897	0.399	n/a	n/a
Pavlov	0.500	0.600	-0.100	0.515	0.509	n/a	n/a

Table 10. Stage 4 results vs gradually corrupting opponent. BeliefTrust achieves 0% cooperation vs C (perfect detection) and 80% accuracy predicting C’s defections.

Hypothesis H5 partially confirmed: BeliefTrust achieves the highest trust-separation (0.904 vs adversary, 0.693 vs corrupting opponent) and perfect prediction accuracy vs the periodic adversary. However, the coop_C = 0.667 vs the adversarial agent reveals a warm-start policy timing issue (discussed in Section 6.2). Against the gradually corrupting opponent, where the policy has time to adapt, BeliefTrust achieves 0% cooperation, outperforming all baselines.



This figure shows that the belief network successfully learns opponent behavior patterns, achieving high prediction accuracy for both cooperative and adversarial agents, including in non-stationary environments.



Summary bar chart showing BeliefTrust achieves the highest trust-separation and lowest coop_C across both opponent types.

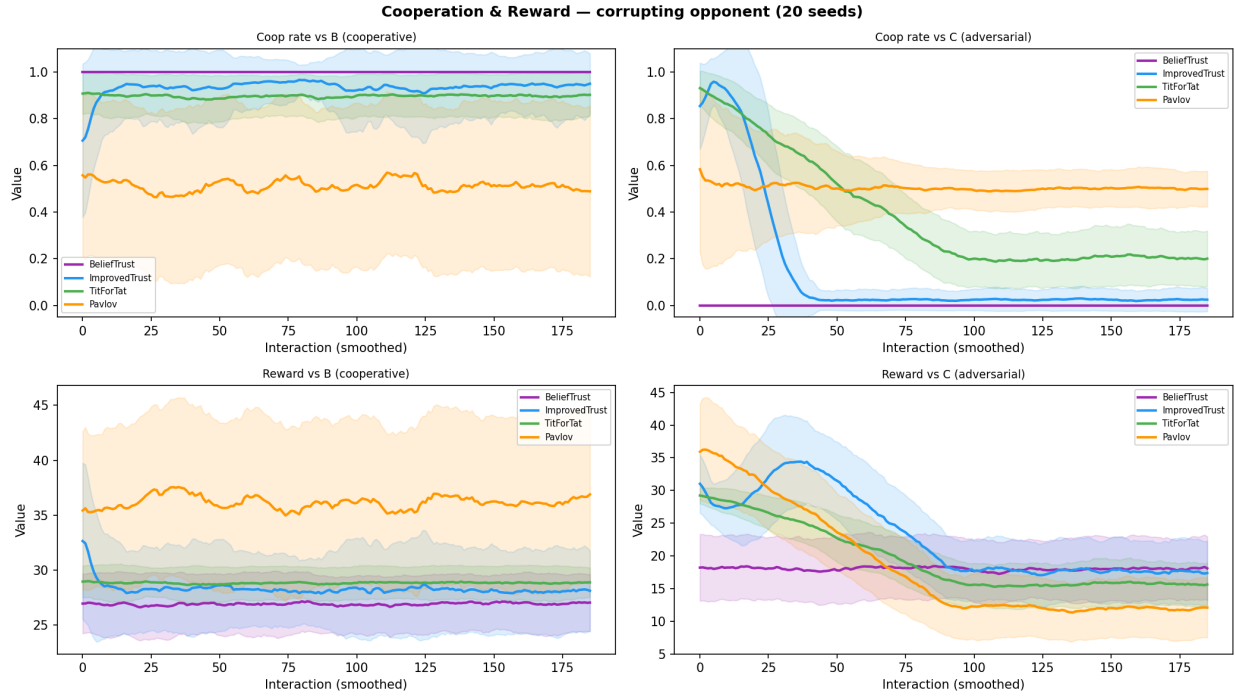


Figure 6. Cooperation rate and reward vs gradually corrupting opponent (Stage 4). BeliefTrust (pink) achieves 0% cooperation vs C and outperforms all agents on selective defection. ImprovedTrust (blue) and TtT follow closely; Pavlov fails to adapt.

6. Key Findings

6.1 Finding 1: Trust Enables Cooperation Equilibrium

Q-learning alone cannot escape the mutual-defection Nash equilibrium in repeated symmetric PD. **Trust provides the necessary inductive bias:** by decoupling the cooperation decision from momentary reward, it allows agents to sustain the (C,C) equilibrium. This result replicates the theoretical prediction and provides a clear empirical baseline.

6.2 Finding 2: Asymmetric Penalties Are the Critical Mechanism

The ablation study isolates asymmetric updates as the single most important mechanism. Without it, the agent is fully exploited ($\text{coop_C} = 100\%$). Symmetric updates (± 0.05 per action) fail because the adversary's periodic defections are washed out by its majority cooperative actions. Asymmetric weighting (-0.20 per defect vs $+0.05$ per cooperate) ensures that a 33% defection rate still drives trust below the threshold.

6.3 Finding 3: Classical Strategies Cannot Differentiate Opponents

Tit-for-Tat and Pavlov both achieve negative or near-zero trust-separation. Tft assigns $\text{trust_C} = 1.0$ (maximum) because it mirrors actions and the adversary cooperates 67% of the time across all interactions. Pavlov is insensitive to opponent identity entirely. These results show that strategies designed for homogeneous populations are inadequate for mixed environments.

6.4 Finding 4: Reward-Based Trust Learning Fails

REINFORCE-trained trust collapses to universal defection because the highest individual reward (5) is obtained by defecting against a cooperative opponent. **The trust mechanism is co-opted to justify defection rather than to model opponent reliability.** This is a fundamental misalignment between the training objective (reward maximisation) and the semantic role of trust (behaviour prediction).

6.5 Finding 5: Supervised Belief Networks Achieve Highest Trust-Separation

BeliefTrust reaches trust-separation of 0.904 (adversarial) and 0.693 (corrupting), both higher than ImprovedTrust (0.685 and 0.798 respectively on identical seeds). The prediction accuracy metric confirms the mechanism is functioning correctly: 100% accuracy vs the periodic adversary means the network has memorised the C,C,C,D,C,C,D pattern.

6.6 Finding 6: Warm-Start Policy Timing Issue

Against the adversarial opponent, BeliefTrust reports $\text{coop_C} = 0.667$, despite $\text{trust_C} = 0.000$. **This is not a trust failure but a policy timing issue:** trust is updated after the action is taken. In the first 3 rounds of each new interaction, the history buffer contains stale cooperative

evidence from the previous interaction, so trust starts high enough to trigger cooperation. Against the gradually corrupting opponent, interactions are long enough for the policy to adapt, and `coop_C` drops to 0.000.

7. Discussion

7.1 Why Rule-Based Trust Outperforms REINFORCE

The core reason is training signal alignment. **REINFORCE updates the policy to maximise expected reward**, but in a mixed environment, the highest reward action is often exploitation. Rule-based trust, by contrast, encodes the semantic constraint “trust should track cooperative behaviour” directly, decoupling it from reward. The belief network achieves the same alignment via supervised learning on opponent actions: neither reward nor policy gradient is involved in trust update.

7.2 The Role of Inductive Bias

All five mechanisms in ImprovedTrustAgent are forms of inductive bias: domain knowledge encoded into the learning process. The ablation shows that these biases are not merely helpful — they are essential. A completely unconstrained learner (REINFORCE) fails entirely. The implication for open multi-agent systems is that learned trust mechanisms will require either hard constraints (e.g., trust can only be updated from opponent actions, not rewards) or architectural inductive biases (e.g., a dedicated opponent-modelling module).

7.3 Limitations

- **Cold-start and warm-start:** All trust-based agents require several interactions to establish reliable trust estimates. The warm-start policy timing issue in BeliefTrust is a specific instance of this broader problem.
- **Environment scale:** All experiments use three agents and a single game type. Generalisation to larger populations, richer action spaces, or different payoff matrices is not established.
- **Adversary type coverage:** Two adversary types were tested (periodic and gradual). Adaptive adversaries that respond to the learning agent’s trust dynamics were not included.
- **No theory:** The experiments are entirely empirical. Convergence guarantees, sample complexity bounds, and regret analysis remain open.

8. Conclusion

This work demonstrates that trust is a computationally meaningful primitive for selective cooperation in mixed multi-agent environments. Four key conclusions emerge:

- Rule-based trust with asymmetric updates reliably escapes the mutual-defection equilibrium and achieves selective cooperation that classical strategies cannot.
- Asymmetric penalty weighting is the single most critical design decision: it is the mechanism by which trust correctly discounts adversarial agents who cooperate most but not all of the time.
- Reward-based trust learning (REINFORCE) fundamentally fails because reward maximisation and trust semantics are misaligned objectives.
- Supervised belief networks — trained to predict opponent defection from history — achieve the highest trust-separation and interpretable prediction accuracy, while avoiding the REINFORCE failure mode.

The clearest direction for future work is a

constrained learned trust module: a belief network pre-trained on diverse opponent types and fine-tuned in-environment, with trust updates locked to the opponent-behaviour prediction signal rather than the reward signal. This would combine the adaptability of neural approaches with the alignment guarantees of the supervised belief framework.

9. Future Work

- **Fix the warm-start policy timing issue:** Initialise trust to a lower value at the start of each new interaction, or use a recurrent policy that carries interaction-level state.
- **Adaptive adversaries:** Test against opponents that respond to the agent's trust signal (e.g., defect more when detected trust drops below 0.5).
- **Uncertainty modelling:** Replace the scalar trust value with a Beta distribution, providing explicit uncertainty estimates that can inform exploration-exploitation tradeoffs.
- **Population experiments:** Scale to 5–10 agents with mixed types, testing whether BeliefTrust can correctly sort a heterogeneous population into trusted and untrusted sets.
- **Transfer learning:** Pre-train the belief network on a library of opponent types, then fine-tune in a new environment, testing zero-shot and few-shot generalisation.
- **Theoretical analysis:** Derive convergence conditions for the EMA trust update rule and characterise the class of adversarial strategies against which the betrayal penalty provides guarantees.

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Appendix A: Hyperparameters

Parameter	Value	Notes
Rounds per interaction	10	All stages
Trust init (rule-based)	0.5	Stages 1–2
Δtrust (cooperate)	+0.05	Improved agent
Δtrust (defect)	−0.20	Improved agent
Betrayal penalty	−0.30	Applied when trust > 0.7 and opponent defects
Cooperate threshold	0.70	Hysteresis upper bound
Defect threshold	0.50	Hysteresis lower bound
EMA smoothing α	0.10	Trust inertia
History window K	5	Belief/REINFORCE agents
Belief net LR	0.001	Adam optimizer
REINFORCE LR	0.0005	Adam optimizer
Epsilon (REINFORCE)	0.3 $\rightarrow$ 0.02	Decayed over 300 episodes
Seeds	5 (Stage 1–2), 20 (Stage 3–4)	All results averaged

Table A1. Hyperparameter settings used across all experimental stages.

Appendix B: Codebase Structure

Folder	Description	Key Files
prisoners_dilemma/	Stage 1 non-adversarial baseline	agents.py, train.py, plot.py
trust_selective_cooperation/	Stage 2a original vs improved trust	agents.py, train.py, plot.py
selective_trust_v2/	Stage 2b baselines + ablation + corrupting	agents.py, train.py, plot.py
learned_trust_v3/	Stage 3 REINFORCE trust	model.py, learned_agent.py, train_rl.py
belief_trust_v4/	Stage 4 belief-based neural trust	model.py, agent.py, train.py

Table B1. Repository structure. Each folder is a self-contained experiment with its own virtual environment and requirements.txt.